

What is a Variable?

- A variable is a container that stores a value.
- It always starts with \$.
- PHP decides the type automatically (dynamic typing).
- Variables are case-sensitive.

Rules for Naming Variables

- Must start with \$ then a letter or underscore.
- Cannot start with a number.
- Cannot contain spaces.
- \$Name and \$name are different.

PHP Data Types

- String → "Taha"
- Integer → 20
- Float → 3.14
- Boolean → true or false
- Array → ["red", "blue"]
- Object → created from a class
- NULL → empty value

Variable Scopes

Local: created inside a function, used only there.

```
function test() {  
    $x = 10;  
}
```

Global: created outside, not accessible inside unless using global.

```
$x = 10;  
function test() {  
    global $x;  
}
```

Static: remembers its value between function calls.

```
function counter() {  
    static $c = 0;  
    $c++;  
    echo $c;  
}
```

Variable Variables (\$\$)

This means: "use the value of a variable as another variable name".

```
$a = "name";  
$$a = "Taha";
```

This creates: `$name = "Taha";`

Another example:

```
$var = "x";  
$x = 10;  
echo $$var; // 10
```

Superglobal Variables

These are available everywhere:

- `$_GET` → URL parameters
- `$_POST` → form data
- `$_SERVER` → server info
- `$_SESSION` → session data
- `$_COOKIE` → cookies
- `$_FILES` → uploaded files
- `$_REQUEST` → GET + POST
- `$_ENV` → environment variables

Constants

Two ways to declare constants:

```
define("PI", 3.14);  
const VERSION = 1;
```